

Notes on the Code: Demographic Transition and Fiscal Sustainability

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Class **HouseholdOLG()** - Household Block

The class **HouseholdOLG()** contains all the functions relates to the household block of the model.

Instances

The instances of the class are:

- **a_grid**: The asset grid. Is an exogenous parameter
- **z_grid**: The productivity grid. Is an exogenous parameter
- **Pi**: The transition matrix for idiosyncratic productivity of households. This is determined by the function `markov_chain_ar1()`, and is exogenous
- **T & Tref**: Number of periods lived & retirement age. Are an exogenous parameter
- **inputs**: dictionary of calibrated parameters. All of this calibration is exogenous

All of the following variables are determined endogenously

- **outputs**: class which returns the outputs solved in the bellman equation
- **pol_interp**: Convex combination of weights to preserve the mass of assets given endogenous grid choice
- **EVa**: Expected marginal value of assets to the household
- **g**: Age, productivity and wealth distribution
- **aggregates**: Dictionary with aggregate variables

Functions

The functions of the class are:

- **bellman_iteration**: This function solves the value function backwards by iterating period after period from the maximum age T_{ret} . Start with the $\mathbb{E}[V_a]$ being zero for agents at retirement (they die with certainty), use the FOC to obtain an initial value for consumption

$$c_{FOC}(a, z, j) = (u')^{-1} (\beta(1 - p_j) \mathbb{E}[V_a(a', z', j + 1) | a, z, j]) \quad (1)$$

we then take the c_{FOC} object and solve for the consumption policy function using the endogenous grid point method. Notice a very important thing: **The marginal utility of assets $\mathbb{E}[V_a(a', z', j + 1) | a, z, j]$ is decreasing in future assets a' . Therefore, c_{FOC} is increasing in the quantity of future assets.**

After obtaining the consumption policy function at maximum age, $c(a, z, T_{ret})$, we solve for the marginal value of assets $V_a(a, z, T_{ret})$ and use this to iterate backwards and solve for other periods. We obtain the vector of indexes and weights of the linear interpolation

- **update_distribution**: This function calculates the stationary-distribution of assets-income-cohorts $g(a, z, j)$ from the initial distribution of newborns. By hypothesis, we assume that the newborn distribution $g(a, z, 0) = \delta(a - 0, z - med(z))$ is dirac at zero assets and median income, and iterate forward from this distribution to obtain the distribution for other cohorts. To do such, we do a loop in which, for each age j and point in the grid (z_i, a_k) , we use the policy functions to send mass to assets next age j (z_i, a'_k) . Afterwards, use the transition matrix to update the idiosyncratic income shock z to z' , arriving to the next age in the distribution. We do this until the maximum age before certain death.

Obs: In the case of using this function for calculating the transition path of MIT shocks, we use the g_0 argument in the function to start from the previous distribution in the initial period.

- **policy_ss**: Take policy functions from the Bellman Equation
- **distribution_ss**: Take distribution from the update_distribution function
- **steady_state**: Set the instances of the class as the steady state objects calculated via the bellman iteration and the update_distribution
- **mit_shock**: The MIT shock function has the "Mean-Field Game" kind of blocks, with (i) a block for calculating the value function at the new steady state - terminal condition $\mathbb{E}[V_a]$; (ii) a block for iterating backwards the value functions and solving for the policy functions in time; and (iii) a block for iterating the distribution from the initial steady state forward by using the policy function solved in block (ii). However, this function only calculates this path of distribution/aggregates/policy functions.

Other Functions

These functions are out of the household block, and connect that block with other blocks of the model - constituting the general equilibrium.

- **solve_steady_state:** This function is responsible for computing the steady state, connecting the household block with the rest of the model. Define the error function:

$$F(z) = \begin{bmatrix} Y - Y(z) \\ r - r(z) \\ w - w(z) \\ G + (r - g^c)B - T(z) \\ (1 + g^c)Transfer - \Psi_t \end{bmatrix}$$

where $z = (A, \beta, w, \tau, Transfer)$ is the vector of equilibrium objects of the model. To find z^* which clears all markets $F(z^*) = 0$, we use a solver.

Heterogeneous Agents Utils

- **solve_euler_one_asset:** We solve for the next period asset policy function $a'(a, z, j)$ and consumption policy function $c(a, z, j)$ from the Bellman Equation consumption first order condition. From the FOC of the bellman equation above, we take each point on the grid state space (z_i, a_k) and iterate with all the possible choices of assets next period a'_l to find implied consumption

$$c_{here} = \text{budget}(z_i, a_k) - a'_l, \forall l \in \{1, \dots, N_a\} \quad (2)$$

By using equations (1) and (2) together with the Intermediate Value Theorem, there must exist an a^* such that these equations are alike.

$$c_{here}(a) - c_{FOC}(a^*) = 0$$

take the value in the grid a_{iap} which is closest to this a^* (that is, the first value where $c_{FOC}(a_{iap}) > c_{here}(a_{iap})$) and we make a linear interpolation between a_{iap-1} and a_{iap} to solve for the asset policy function $a'(a, z, j)$. This function also returns the index of the asset policy function, based on the point in the grid a_{ind}

- **update_policy_1a:** Function which takes the previous distribution $g_t(a, z)$ for a given cohort of age j to a mid-object which is the distribution next period conditional on no changes in the productivity $g_{mid,t}(a, z)$. It uses the asset policy function and interpolation weights to distribute the mass of $g_t(a, z)$ to $g_{mid,t}(a, z)$.
- **update_exogenous_1a:** This function takes the object $g_{mid,t}(a, z)$ and uses the rouwenhorst discretization matrix to distribute idiosyncratic income shocks, obtaining the next period distribution $g_{t+1}(a, z)$